

Rethinking Statistical Significance in Follow-On Damages Litigation

How Severity Analysis Can Help Courts Assess Econometric Evidence

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1 Introduction

- 1.1 As the EU Damages Directive marks its tenth anniversary, private enforcement of antitrust law has become an indispensable complement to public antitrust enforcement in the European Union. Courts across Member States are now routinely called upon to determine and quantify damages arising from infringements such as cartels, a task that increasingly requires sophisticated economic and statistical analysis. In such cases, economic experts for both claimants and defendants typically submit regression models estimating the overcharge attributable to a competition law infringement. These estimates, however, are inherently subject to statistical uncertainty, which is commonly assessed through tests of statistical significance.
- 1.2 Interpreting the outcomes of these statistical tests has therefore become a crucial task for courts and authorities. Recent judgments highlight the increasingly prominent role of statistical significance in the assessment of cartel damages claims. A review of case law reveals, however, that courts often struggle to effectively use empirical evidence in the judicial decision process. In this article, we argue that this difficulty arises partly from conceptual misunderstandings, but more fundamentally reflects the limitations of conventional statistical tools. Statistical significance is designed to answer a binary question — usually whether an effect can be distinguished from zero — not the question courts ultimately face: how large is the effect, and how confidently can that magnitude be inferred from the data?
- 1.3 In the following sections, we first unpack statistical significance and explore how it works. We then introduce severity analysis — a framework developed by Deborah Mayo and Aris Spanos — as a way to assess what specific claims about effect magnitude the data can and cannot support. The article concludes by proposing severity analysis as a methodological response to this problem.

2 Statistical significance and its use in the assessment of cartel damages

- 2.1 Statistical significance or insignificance is the outcome of a statistical hypothesis test. Such a test assesses whether the observed evidence — in this case the estimated price effect — is sufficiently inconsistent with a previously specified hypothesis, the so-called “null hypothesis”, to justify the rejection of that hypothesis, given the sampling uncertainty of the estimate. To that end, the test evaluates, under the assumption that the null hypothesis is true, the probability of observing an effect at least as far away from the null hypothesis as the one observed in the data. In other words, it assesses whether the observed effect is larger than could plausibly be explained by random variation or statistical noise.

- 2.2 Depending on whether that probability is sufficiently small or not, the test leads to one of two outcomes: If the probability is sufficiently small, the null hypothesis is rejected, and the estimated effect is labelled “statistically significant”. Otherwise, the null is not rejected and thus the estimated effect is labelled “not statistically significant”.
- 2.3 The threshold for what counts as “sufficiently small” is chosen to suit the case at hand. Conventionally, that probability is measured against one of three possible “significance levels”, often indicated with asterisks: 1% (***), 5% (**), and 10% (*). In economics, we typically label an estimate “significant” if the probability of observing this estimate is smaller than 5% under the null hypothesis. That is, we employ the 5% significance level.

How does it work?

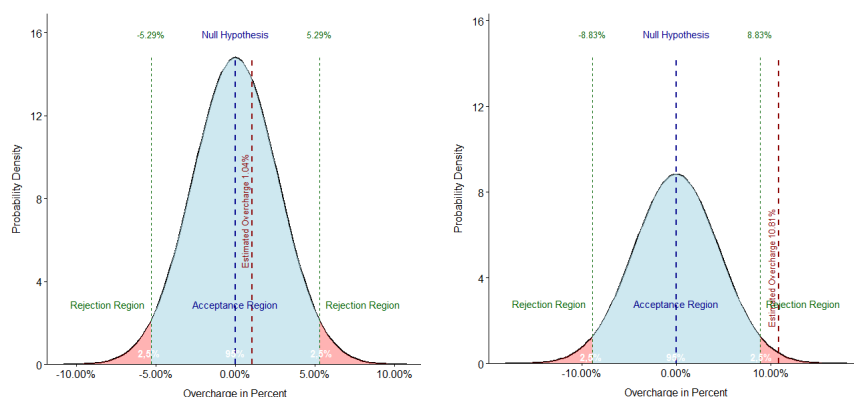
- 2.4 Any empirical estimate is subject to uncertainty because the observed data represent only one possible realisation of the underlying unknown real-world process that generated them. Therefore, even if the true price effect of an infringement was exactly 8%, the observed data would yield estimates that fluctuate around 8% but would almost never equal 8% exactly. The same reasoning applies absent a price effect: the estimated effect will typically be close to zero but not exactly zero. A hypothesis test therefore asks whether the observed estimate is far enough away from the value stated in the null hypothesis that random variation alone is no longer a plausible explanation for the observed estimate.
- 2.5 To do so, the test compares two rival hypotheses.
- The **null hypothesis** usually states that there is no effect – in this case, that the true price effect is zero.¹
 - The **alternative hypothesis** always states the opposite of the null hypothesis, that is, it conventionally states that the true effect is not zero. Thus, the alternative hypothesis covers a whole range of possible values rather than one specific number: any price effect different from zero falls under the alternative hypothesis. A two-sided test therefore treats both sufficiently positive and sufficiently negative estimates as inconsistent with the null hypothesis.
- 2.6 Figure 1 illustrates this logic using two examples. In both panels, the bell-shaped curve shows the distribution of estimates that would be expected if the null hypothesis were true – here, if there was no overcharge due to the infringement, i.e. the true effect was zero. The bell-curve is therefore centred around the null hypothesis of no overcharge. The width of each bell-curve reflects the degree of sampling uncertainty associated with the respective estimate. The area under each curve represents probability: it shows how likely different estimates would be if the null hypothesis were true. In the figure, 95% of that area lies in the central region around zero, meaning that estimates in this range are treated as sufficiently consistent with the null hypothesis that it is not rejected. The remaining 5% lies in the two tails, with 2.5% in each tail, adding up to the pre-defined significance level of 5%. These tail areas are therefore called the rejection regions. They contain estimates that would be so unlikely if the null hypothesis were true that the null hypothesis is rejected. If the estimate falls in one of those regions, the result is labelled statistically significant and treated as

¹ The null hypothesis of no price effect is the most common null hypothesis in damages quantification, because the null hypothesis is typically a statement about the real world that the researcher seeks to test and, if justified by the data, to reject (see the next subsection). More generally, however, the null hypothesis need not be limited to no effect. It can be any statement about a parameter or effect of interest. The logic of the testing approach described in this paper would apply in the same way.

supporting the alternative hypothesis. Under the null hypothesis, less extreme estimates fall into the blue area in Figure 1, also labelled the acceptance region.

- 2.7 It is essential to note what such a significance test is actually designed to measure. The bell-curves, the rejection regions, and the 5% threshold are all defined under the assumption that the null hypothesis is true. **Statistical significance is therefore a statement about how unusual the observed estimate would be if the null hypothesis were true. It is not a statement about the probability that the null hypothesis, or any competing hypothesis, is true given the data.**

Figure 1: Statistical hypothesis test against the null hypothesis of no overcharge for different estimates



Notes: [XX]

Source: Compass Lexecon analysis based on simulated data.

- 2.8 The two panels in Figure 1 differ in the amount of uncertainty surrounding the estimate reflected in the different width of the bell-curves. In the left-hand panel, the number of observations is larger, so the distribution is narrower and the estimated overcharge of 1.04% is more precise. Nevertheless, the observed estimate remains within the central region and is not statistically significant. In the right-hand panel, the number of observations is smaller, so the distribution is wider and the estimate is less precise. Here, the observed estimate of 10.81% falls in the rejection region and is statistically significant. The figures therefore show that statistical significance depends not only on the size of the estimate – and thereby the deviation from the null hypothesis – but also on how precisely it is measured.

Box 1: A loose analogy for statistical significance testing

A struggling football club appoints a new coach and wants to know whether the team is genuinely performing better:

- The coach's unobserved true impact on the team represents the **true effect** we are trying to detect. The number of matches played, and how informative those matches are, represent how much information the test has available. The capacity of the test corresponds to how informative the observed run of matches is for isolating the coach's true effect, taking into account the overall noise in match outcomes and their comparability to the matches under the previous coach, for example in terms of fixture difficulty, weather and refereeing decisions, and other random factors affecting individual match results.

- b. We know that match results are variable. Some variation is systematic and explainable – such as opponent strength, home/away fixtures, or injury absences – while some is harder to predict such as referee decisions, lucky deflections, or weather. The same applies to prices in damages quantification. In a regression analysis, we try to account for observable systematic factors; what remains is the noise against which the estimated effect has to be assessed.
- c. If the club observes a clear improvement over enough matches, it may be reasonable to conclude that the new coach is having some real positive effect on the team's performance. That is analogous to a statistically significant result: the improvement is too pronounced to be plausibly explained by random variation alone.
- d. But if only a few matches have been played, or if the match outcomes are generally very noisy, even a genuinely better coach may fail to produce results that stand out clearly. That is analogous to a statistically insignificant result: it may be insignificant not because there is no improvement, but because the evidence is too noisy or too limited to detect it clearly.

Interpreting statistical significance in follow-on damages cases

- 2.11 Now that we have established how statistical significance testing works, it is worth explicitly noting the challenges courts face when interpreting statistical significance in damages cases.
- 2.12 Based on our discussion in the previous section, the outcome of a statistical significance test can be:
 - a. A **statistically significant result**, which means that, if the null hypothesis were true, a deviation from the null hypothesis of this size or more extreme would be sufficiently unlikely for the null hypothesis to be rejected in favour of the alternative hypothesis.
 - b. A **statistically insignificant result**, which means that the observed evidence is not strong enough to reject the null hypothesis. It does not prove that there is no effect, nor does it say anything about how consistent the result is with an alternative, economically relevant hypothesis or about the magnitude of the effect itself. It means only that, given the degree of uncertainty associated with the estimate, the possibility remains that the observed result reflects random variation rather than a real effect.
- 2.13 Against this background, courts and authorities that rely on statistical significance to assess empirical evidence risk two major interpretive errors:
 - a. **The fallacy of acceptance:** Treating a non-significant result (failure to reject the null) as proof that the null is true. A non-significant finding only shows that the data do not provide sufficient evidence against the null; it says nothing about the probability that the null is correct. The absence of evidence for an effect is not evidence of the absence of an effect. A test may fail to reject the null simply because the number of observations is limited or the data are highly variable. In such circumstances, the evidence is inconclusive rather than confirmatory of the null hypothesis.
 - b. **The fallacy of rejection:** Treating a significant result (rejection of the null) as confirmation that the estimated effect is accurate or economically meaningful. A significant result only indicates that an estimate as extreme as the observed estimate would be unlikely if the null were true. It does not establish the size of the true effect. For example, a statistically significant overcharge estimate of 12% does not, by itself, establish whether the true overcharge is close to 12%, materially lower, or materially higher.

- 2.14 Standard statistical significance therefore offers little help to a court that must rule on the size of the effect. Faced with this limitation, courts have either reached for the *broad axe*² – disregarding statistical evidence altogether – or have selectively discounted statistically insignificant results when evaluating the magnitude of the alleged harm, perhaps to avoid the fallacy of acceptance.³
- 2.15 This difficulty is particularly acute where statistically insignificant estimates are presented alongside statistically significant expert evidence. If an insignificant result is treated as equivalent to a finding of no effect, it becomes difficult to reconcile that result with other evidence pointing to a positive effect. One possible judicial response has therefore been to attach little or no weight to statistically insignificant analyses when estimating the amount of harm.⁴ However, this response risks replacing one error with another: an insignificant result should not be treated as proof of zero harm, but neither should it be ignored altogether, since it may still provide relevant information about the plausible magnitude of the effect, depending on the precision of the estimate and the range of values that remain compatible with the data.
- 2.16 The core shortcoming is this: significance testing produces a binary verdict (reject or do not reject the null of no effect) but says little about the **plausibility or magnitude of the underlying effect**. Courts, however, need to assess not whether an effect exists but how large it is and how confidently that magnitude can be inferred from the data. That requires a framework geared to specific effect sizes rather than to the null hypothesis alone. The concept of *severity*, introduced in the next section, provides such a framework.

Box 2: A loose analogy for misleading interpretations of the outcomes of a statistical test

We can continue with our football coach analogy to illustrate the two most common mistakes in the interpretation of statistical significance.

a. The fallacy of rejection:

- If the new coach oversees a clear winning run such that we conclude that this cannot be explained by mere luck given the overall variation in outcomes (i.e., we reject the null hypothesis), it may be reasonable to conclude that the improvement is real
- But it would be wrong to conclude from that alone that the coach must be outstanding, or that the team has improved enough to achieve a specific level of improvement such as increasing the average probability of winning by X%.
- Likewise, a statistically significant result may justify rejecting the null hypothesis, but it does not by itself show that the effect is large or practically important (e.g., the increase in the winning percentage may still not be enough for the team to avoid relegation at the end of the season).

b. The fallacy of acceptance:

- Suppose the new coach does not produce obviously better results over the first couple of matches. It would be wrong to conclude from that alone that the new coach has had no positive effect on the team's performance at all.

² [Add footnote referencing UK cases such as Royal Mail.]

³ [Add footnote referencing German cases where the court only included significant findings.]

⁴ [Add footnote referencing German cases where the court only included significant findings.]

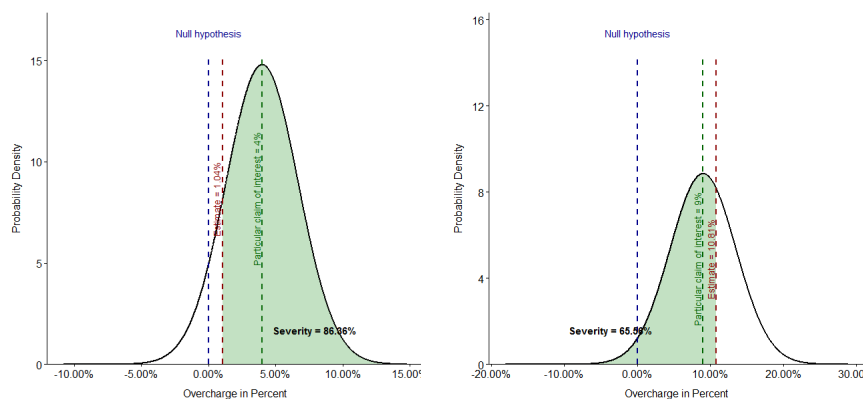
- The run of matches may simply have been too short, too difficult, or too noisy to reveal it clearly and/or the level of improvement may simply be too small to be measured under these circumstances.
- Likewise, a statistically non-significant result does not prove that there is no effect. It may only show that the data were too limited or too noisy to detect it clearly.

3 Introducing Severity Analysis

- 3.1 While standard hypothesis testing provides a binary “yes/no” answer to the question whether the observed effect is unlikely under the null hypothesis, a severity assessment allows us to rule out effect sizes of particular magnitudes.⁵
- 3.2 The reasoning builds directly on the result of a standard test. When the null is not rejected, the outcome may simply reflect that the test had little capacity to detect a real effect — in which case non-rejection is only weak evidence that the effect is absent. Severity addresses this limitation by asking a different question: *which effect sizes are inconsistent with the observed estimate, given the capacity of the test?*
- 3.3 Consider a hypothetical cartel case in which a regression yields a statistically insignificant overcharge estimate of 1.04% (left panel of Figure 1). Concluding from this that there was no overcharge would commit the fallacy of acceptance. A severity test instead asks: does the evidence support the claim that the true overcharge was no larger than some level the court regards as relevant — say, 4%?
- 3.4 To answer this, we calculate the probability of obtaining an estimate larger than 1.04% under hypothesized alternative overcharges (e.g. 4%, 5%, 10%, or any other value of interest):
- i. If that probability is high — i.e., had the true overcharge been 4%, the test would very likely have produced an estimate above the actually observed 1.04% — then the failure to observe such an estimate is strong evidence that the true overcharge is below 4%. The claim “overcharge $\leq$ 4%” passes with high severity.
 - ii. If that probability is low, the test lacked the capacity to distinguish a 4% overcharge from what we actually observed. The claim “overcharge $\leq$ 4%” passes only with low severity, and a true overcharge of 4% or more cannot be ruled out.
- 3.5 The intuition is simple: **a test that had the capacity to detect a larger discrepancy, but did not, provides evidence against the existence of a larger discrepancy.** Certain overcharge magnitudes can therefore be ruled out even when the point estimate is statistically insignificant in the conventional sense.
- 3.6 Figure 2 illustrates this graphically using the bell-curves introduced in Figure 1:
- a. The left panel depicts the statistically insignificant estimate of 1.04%. Severity asks: how likely would it have been to obtain an estimate above 1.04% if the true overcharge had been 4% or larger? Graphically, this is the area under the bell-curve to the right of 1.04%. In our example, that probability exceeds 86%. The test therefore had a high capacity to produce an estimate above 1.04% had the true overcharge been 4% or larger; the fact that it did not warrants the conclusion, with high severity, that the true overcharge is below 4%.

⁵ [Add footnote mentioning Mayo and Spanos as authors of the concept.]

Figure 2: Severity test against a particular claim of interest for different test outcomes



Notes: [XX]

Source: Compass Lexecon analysis based on data from

- b. The right panel applies the same logic to a statistically significant result. Here, severity asks how likely it would have been to obtain an estimate *smaller* than the observed 10.81% if the true overcharge had been only 9%. If that probability is high, the fact that the observed estimate is larger provides strong evidence, with high severity, that the true overcharge exceeds 9%.

Box 3: A loose analogy for severity

Returning to the football coach analogy: statistical significance tells us whether the observed performance improvement is clear enough, judged against the performance under the previous coach, to conclude that what we see is unlikely to be mere luck. Severity asks a further question: how capable was the run of matches of revealing an improvement of the magnitude we are interested in, had one really existed?

a. Severity when the result is statistically insignificant:

- Suppose the early results show no clear improvement. That does not by itself prove that the coach has made no difference, as the matches may have been too few, or the best player may have been injured.
- The club can still ask whether the run of matches was long and informative enough that a genuine improvement of a particular size would probably have shown up in the records.
- If so, the absence of a clear improvement is reason to rule out improvements of *that* size – even if smaller improvements cannot yet be excluded.
- If not, the evidence is not strong enough to rule out an improvement of that size.
- Likewise, a statistically insignificant result may still allow us to rule out an effect of a particular size, if the test had enough capacity to detect such an effect had it existed.

b. Severity when the result is statistically significant:

- Suppose the new coach does produce clearly better outcomes. That may justify concluding that the team has improved.
- But the club can still ask whether the evidence is strong enough to support a more specific claim — for example, that the coach improved the team by more than a particular threshold.
- If a smaller improvement would most likely have looked less impressive than what was actually observed, the stronger claim passes with high severity.
- If, however, the observed results could easily have arisen even with an improvement below that threshold, the stronger claim has low severity.
- Likewise, a statistically significant result may support not only the conclusion that there is some effect, but also more specific claims about the size of the effect, provided the test would probably have revealed it if those claims were false.

In cartel damages, the same logic applies in both directions: a statistically insignificant estimate can still rule out overcharges above a certain size, and a statistically significant estimate can support claims that the true overcharge exceeds a particular threshold of interest.

- 3.7 Severity therefore allows courts to extract information about the plausible magnitude of the effect from both statistically insignificant and statistically significant results: in the former case by ruling out larger effects, and in the latter by supporting claims that the true effect exceeds a particular magnitude of interest. Severity analysis thereby addresses both fallacies by shifting the focus from whether a null hypothesis can be rejected to assessing how strongly the estimated overcharge speaks against specific effect-size claims. Importantly, the actual estimated overcharge matters for

the severity assessment, whereas the binary decision in standard statistical significance testing treats estimates within the rejection region, or within the non-rejection region, in the same way.

3.8 In the next section, we demonstrate how the logic of severity testing can be operationalised in cartel damages cases.

4 How severity analysis can help in practice

4.1 As we have shown above, while neither a statistically insignificant nor a statistically significant overcharge estimate can prove the absence of damage (i.e., an overcharge of exactly zero) or the existence of an effect of a particular magnitude, every estimate – statistically significant or not – carries information about the range of plausible effect sizes within the severity framework.^{6,7}

4.2 Returning to the statistically insignificant estimate of 1.04%: the practical question is to identify the smallest hypothetical true overcharge for which an estimate above 1.04% would have been so likely that the failure to observe one effectively rules that magnitude out. Put differently, severity asks how likely it would have been to observe an estimate larger than 1.04% if the true overcharge had in fact been a specific higher value. The more precise the underlying estimate, the smaller the alternative overcharges that can be excluded – so severity also reflects the informational strength of the analysis.

4.3 Concretely, if the probability of observing an estimate above 1.04% under a candidate alternative – say 2%, 3% or 5% – is sufficiently high (e.g. 80%), the result is inconsistent with an effect of that size or larger, because such an estimate was not observed.

4.4 Plotting severity for every candidate alternative overcharge produces the upward-sloping severity curve shown in Figure 3 below.⁸ Two features help read it:

- a. At a hypothetical true overcharge equal to the estimate itself, severity is always 50% (indicated by the green colour). This is intuitive: If the true effect were 1.04%, the probability of estimating an effect above 1.04% is exactly 50%.
- b. The further right we move along the horizontal axis, the higher the probability of observing an estimate above 1.04% – and therefore the higher the severity with which that magnitude can be ruled out.

4.5 Applying an 80% severity threshold, the analysis shows that any overcharge above 3.3% can be ruled out given the observed estimate of 1.04%.⁹ This is illustrated in Figure 3 by the point at which the grey severity curve intersects the vertical line at an overcharge of 3.3%.

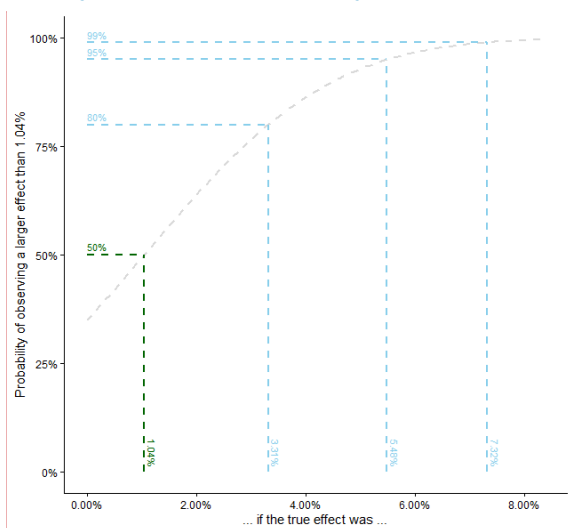
⁶ [In this sense, the severity approach can also be used to reconcile supposedly contradictory evidence. See Inderst Bönisch]

⁷ [Add footnote saying that this can also be derived from a Bayesian approach, referencing the AER Paper from Abadie.]

⁸ [Add footnote saying that an alternative presentation of severity would be the inverse severity curve which is what we have submitted in TRKS. Would nevertheless stick to the official definition for the purpose of the paper.]

⁹ [Add footnote explaining the standard approach of the 80% power level.]

Figure 3: The severity curve for the case of non-rejection



Notes: [Mention that the severity curve for rejection would be downward sloping.]
Source: Compass Lexecon analysis based on data from

Commented [PB1]: I have changed the figures labelling and the 50% example to ave a different color. I hope this helps.

- 4.6 More generally, severity augments the conventional testing logic. Once the significance test outcome, the estimate, and its associated uncertainty are known, severity uses the same estimate and its uncertainty to map *which specific magnitudes* the data can rule out. When the null is not rejected, severity bounds the effect from above, identifying which discrepancies from zero can still be excluded. When the null is rejected, severity bounds the effect from below, identifying the magnitudes the data positively support. Either way, the court is told what the evidence actually shows about the possible *size* of the overcharge, rather than stopping at the binary verdict that the estimate is either statistically significant or statistically insignificant.